

2(a)	$v^2 = u^2 + 2as$ $u^2 = 8.7^2 - (2 \times 9.81 \times 1.5)$	C1
	$u = 6.8 \text{ m s}^{-1}$	A1
2(b)	(magnitude of) force on ball (by ground) equal to force on ground (by ball)	B1
	(direction of) force on ball (by ground) opposite to force on ground (by ball)	B1
2(c)(i)	$(p =) 0.059 \times 8.7 \text{ or } 0.059 \times 5.4$	C1
	change in momentum = $0.059 (8.7 + 5.4)$ $= 0.83 \text{ N s}$	A1
2(c)(ii)	resultant force = $0.83 / 0.091 \text{ or } 0.059 [(8.7 + 5.4) / 0.091]$ $= 9.1 \text{ N}$	A1
2(c)(iii)	$(W =) 0.059 \times 9.81$	C1
	$(W =) 0.58 \text{ (N)}$ force = $9.1 + 0.58$ $= 9.7 \text{ N}$	A1
2(d)	straight line with a positive gradient and starting from a non-zero value of speed at $t = 0$ and ending when $t = T$	B1
2(e)	air resistance increases	B1
	resultant force/acceleration decreases so gradient (of curve) decreases	B1